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EDUCATION & TRAINING

31/08/2022 - 30/11/2025 - BOLOGNA, ITALY

ARTIFICIAL INTELLIGENCE - MASTER OF ENGINEERING - ALMA MATER STUDIORUM – UNIVERSITÀ DI BOLOGNA

Thesis: AI-Driven CT Atlas for Lung Nodules in Cancer Screening

31/08/2006 - 31/12/2008 - HAMADAN, IRAN

PHYSICS , MASTER OF SCIENCE- BU-ALI SINA UNIVERSITY

Thesis: Dark Energy and its Relation to Cosmological Constant.

30/09/2001 - 31/01/2006 - TEHRAN, IRAN

CHEMICAL ENGINEERING & CHEMISTRY , BACHELOR OF SCIENCE- KHARAZMI UNIVERSITY

WORK EXPERIENCE

2015 - 2022 - KARAJ, IRAN

CERTIFIED HIGH SCHOOL PHYSICS TEACHER MINISTRY OF EDUCATION

Planning and creating teaching materials; Preparing and delivering lecture sessions; Assigning assignments to the students; Grade students' assignments and exams and maintaining examination grades; Using teaching methods depending on the students learning abilities

2010 - 2014 - TEHRAN, IRAN

INFORMATION TECHNOLOGY SUPPORT ENGINEER A LOCAL COMPANY

Building network configurations and connections; IP address auditing; Establishing networking environment by designing system configuration and directing system installation; Security of data and network access; Troubleshoot and identify systems and network problems; Configuring and testing systems

PUBLICATIONS

Enhancing Lung Cancer Detection: Segmentation and Classification of CT Images Using U- Net with PretrainedBackbones and BIR Model 2024

The aim of this study is to improve the segmentation and classification of lung regions from CT images of 78 chest cancer patients, focusing on enhancing the accuracy in detecting cancerous tissues. This research evaluates the performance of different U-Net backbone models (VGG16, ResNet50, Xception) in segmentation and employs a novel BIR-enhanced CNN model for classifying lung injury severity.

A U-Net model with three different backbones—VGG16, ResNet50, and Xception—was utilized for lung region segmentation. Preprocessing techniques such as CLAHE (Contrast Limited Adaptive Histogram Equalization) were applied to enhance contrast and image quality, followed by resizing to 128x128 pixels and normalization. For classification, a BIR-enhanced CNN model was employed to assess lung injury severity. The models were evaluated across multiple metrics, including accuracy, recall, F1 score, Intersection over Union (IoU), and Dice coefficient.

Among the models, VGG16 achieved the highest performance, with an accuracy of 0.9836 ± 0.0177, recall of

0.9696 ± 0.0737, F1 score of 0.9363 ± 0.0832, IoU of 0.8893 ± 0.1178, and Dice coefficient of 0.9363 ± 0.0832 in segmentation tasks. For the classification of lung injury severity, the BIR-enhanced CNN model, also utilizing VGG16, achieved a classification accuracy of 97.83%.

This study demonstrates the significant impact of preprocessing on segmentation and classification performance. The U-Net model with the VGG16 backbone not only provides highly accurate segmentation of lung regions but also highlights cancerous areas effectively. The integration of the BIR model further improves classification accuracy, indicating that this combination offers an effective approach for lung cancer detection and diagnosis.

PREPRINT (Version 1) available at Research Square

DenUnet: enhancing dental image segmentation through edge and body fusion 2024

Accurate tooth segmentation is of paramount importance in oral healthcare because it provides critical positional data for clinical diagnosis, orthodontic treatment, and surgical procedures. Despite the widespread use of convolutional neural networks (CNNs) in image segmentation, its limitations in collecting complete global context and long-range interactions are acknowledged. Although vision transformers show promise in understanding larger contextual information, they struggle to manage the spatial complexities of medical images. To tackle these issues, the proposed DenUnet leverages the strengths of both CNNs and transformers. It introduces a dual-branch encoder that simultaneously extracts edge and body information and a double-level fusion module for integrating multi-scale features. To ensure the fine fusing of edge and body information derived from the two mentioned encoders, we propose a local cross-attention feature fusion module to enhance feature fusion with accurate blending losses. Experimental results underscore the superior efficacy of DenUnet in comparison to existing methods. We achieved 95.4% accuracy and 92.7% F1-score on the DNS dataset, which is particularly evident in its ability to adeptly handle irregular boundaries of dental datasets.

GITHUB

GitHub Acount

Links https://github.com/FarhadBayrami

SKILLS

Python | Pytorch | Keras | Tensorflow | Scikit-Learn | SimpleITK | OpenCV | Git | Linux | LaTeX / Microsoft Office

LANGUAGE SKILLS

Mother tongue(s): **PERSIAN**

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1

RECOMMENDATIONS

Professor Stefano Diciotti

Thesis Supervisor

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Dr. Giulia Raffaella De Luca

Thesis Co-supervisor

Email giulia.deluca19@unibo.it